

DW_lzd

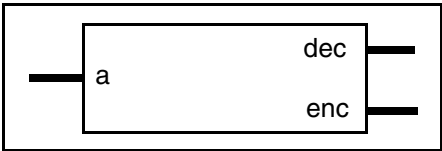
Leading Zeros Detector

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Features and Benefits

- Parameterized word length
- Inferable using a function call

Revision History



Description

DW_lzd contains two outputs, dec and enc. The dec output is a decoded one-hot value of the a input vector assuming there is at least one 1 on a. The output enc represents the number of 0s found (from the most significant bit) before the first occurrence of a 1 from the input port a. All lower order bits (to the right) from the first occurrence of the “1” on the a input port are “don't care.” If no 1 is found and only 0s are present, the resulting value of enc is all 1s and of dec is all 0s.

The output port enc width is automatically derived from the input port width parameter, *a_width*, and is defined as $\text{ceil}(\log_2[a_width]) + 1$ as listed in [Table 1-1](#). Output port dec has the same width as the a input.

Table 1-1 Pin Description

Pin Name	Width	Direction	Function
a	a_width	Input	Input vector
enc	$\text{ceil}(\log_2[a_width]) + 1$	Output	Number of leading 0s in input a before first 1. All 1s if input a is all 0s.
dec	a_width	Output	One-hot decode of input a - All 0s if input a is all 0s.

Table 1-2 Parameter Description

Parameter	Values	Description
a_width	≥ 1 Default: 8	Vector width of input a

Table 1-3 Synthesis Implementations

Implementation Name	Function	License Feature Required
cla	Synthesis model	DesignWare
rtl	Synthesis model	DesignWare

Table 1-4 Simulation Models

Model	Function
DW01.DW_LZD_CFG_SIM	Design unit name for VHDL simulation
dw/dw01/src/DW_lzd_sim.vhd	VHDL simulation model source code
dw/sim_ver/DW_lzd.v	Verilog simulation model source code

Table 1-5 Truth Table (*a_width* = 7)

a(6:0)							enc(3:0)	dec(6:0)						
0	0	0	0	0	0	0	1111	0	0	0	0	0	0	0
0	0	0	0	0	0	1	0110	0	0	0	0	0	0	1
0	0	0	0	0	1	X	0101	0	0	0	0	0	1	0
0	0	0	0	1	X	X	0100	0	0	0	0	1	0	0
0	0	0	1	X	X	X	0011	0	0	0	1	0	0	0
0	0	1	X	X	X	X	0010	0	0	1	0	0	0	0
0	1	X	X	X	X	X	0001	0	1	0	0	0	0	0
1	X	X	X	X	X	X	0000	1	0	0	0	0	0	0

Related Topics

- [Math – Arithmetic Overview](#)
- [DesignWare Building Block IP User Guide](#)

HDL Usage Through Function Inferencing - VHDL

```
library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DW_Foundation.all;
-- If using numeric_std data types of std_logic_arith, uncomment the
-- following line:
-- use DWARE.DW_Foundation_arith.all;

entity DW_lzd_func is
  generic (
    func_a_width : POSITIVE := 8
  );
  port (
    func_a : in std_logic_vector(func_a_width-1 downto 0);
    dec_func : out std_logic_vector(func_a_width-1 downto 0);
    enc_func : out std_logic_vector(bit_width(func_a_width) downto 0)
  );
end DW_lzd_func;

architecture func of DW_lzd_func is
begin

  -- Inferred function of DW_lzd
  dec_func <= DWF_lzd(func_a);
  enc_func <= DWF_lzd_enc(func_a);

end func;

-- pragma translate_off
configuration DW_lzd_func_cfg_func of DW_lzd_func is
for func
end for; -- func
end DW_lzd_func_cfg_func;
-- pragma translate_on
```

HDL Usage Through Function Inferencing - Verilog

```
module DW_lzd_func( func_a, dec_func, enc_func );

parameter func_a_width = 8;

`define enc_width 4 // ceil(log2(func_a_width))+1

// Passes the widths to DW_lzd_function
parameter a_width      = func_a_width;
parameter addr_width   = `enc_width;
`include "DW_lzd_function.inc"

input  [func_a_width-1 : 0] func_a;
output [func_a_width-1 : 0] dec_func;
output [`enc_width-1 : 0] enc_func;

// Function inference of DW_lzd and DW_lzd_enc
assign dec_func = DWF_lzd(func_a);
assign enc_func = DWF_lzd_enc(func_a);

endmodule
```

HDL Usage Through Component Instantiation - VHDL

```
library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DW_Foundation_comp.all;
use DWARE.DWpackages.all;
-- If using numeric_std data types of std_logic_arith, uncomment the
-- following line:
-- use DWARE.DW_Foundation_arith.all;

entity DW_lzd_inst is
  generic (
    inst_a_width : POSITIVE := 8
  );
  port (
    inst_a : in std_logic_vector(inst_a_width-1 downto 0);
    dec_inst : out std_logic_vector(inst_a_width-1 downto 0);
    enc_inst : out std_logic_vector(bit_width(inst_a_width) downto 0)
  );
end DW_lzd_inst;

architecture inst of DW_lzd_inst is
begin

  -- Instance of DW_lzd
  U1 : DW_lzd
    generic map ( a_width => inst_a_width )
    port map ( a => inst_a, dec => dec_inst, enc => enc_inst );
end inst;

-- pragma translate_off
configuration DW_lzd_inst_cfg_inst of DW_lzd_inst is
  for inst
    end for; -- inst
end DW_lzd_inst_cfg_inst;
-- pragma translate_on
```

HDL Usage Through Component Instantiation - Verilog

```
module DW_lzd_inst( inst_a, dec_inst, enc_inst );

parameter a_width      = 8;
parameter enc_width    = 4;  // ceil(log2(a_width))+1

input  [a_width-1 : 0] inst_a;
output [a_width-1 : 0] dec_inst;
output [enc_width-1 : 0] enc_inst;

    // Instance of DW_lzd
    DW_lzd #(a_width)
        U1 ( .a(inst_a), .dec(dec_inst), .enc(enc_inst) );

endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
December 2023	DWBB_202312.0	■ Updated date and version; no technical changes ■ Added this Revision History table and the document links on this page

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